

Quality Benchmark for the AI Tutor

Before-and-after evaluation across 50 simulated chat sessions

Report generated May 02, 2026

HEADLINE RESULT

98% fewer quality issues

From 229 violations across 50 sessions to 5, after one cycle of prompt + guard improvements.

229 -> 5

total quality violations across all 50
sessions

0%

violation rate on 7 of 8 categories with
the new system

432

Maya replies analysed in the after-run
(427 in before-run)

Methodology: identical 50-session schedule (same-day, consecutive-day, sporadic, cold-start, and weekly deep-run patterns) on Qwen 32B via AWS Bedrock, with the same Qwen-tuned prompt set in both runs. The only variable was the post-processing guard layer.

Executive summary

Miss Maya is PeerUp's AI tutor for spoken English. To make her replies feel like a real friend rather than a scripted bot, we ran a structured quality study across 50 simulated user chats and improved the system in two layers.

1. The prompt layer.

We rewrote the prompts that govern Maya's tone, character, opening lines, and memory use - tuned specifically for the Qwen 32B model we run in production.

2. The safety-net layer.

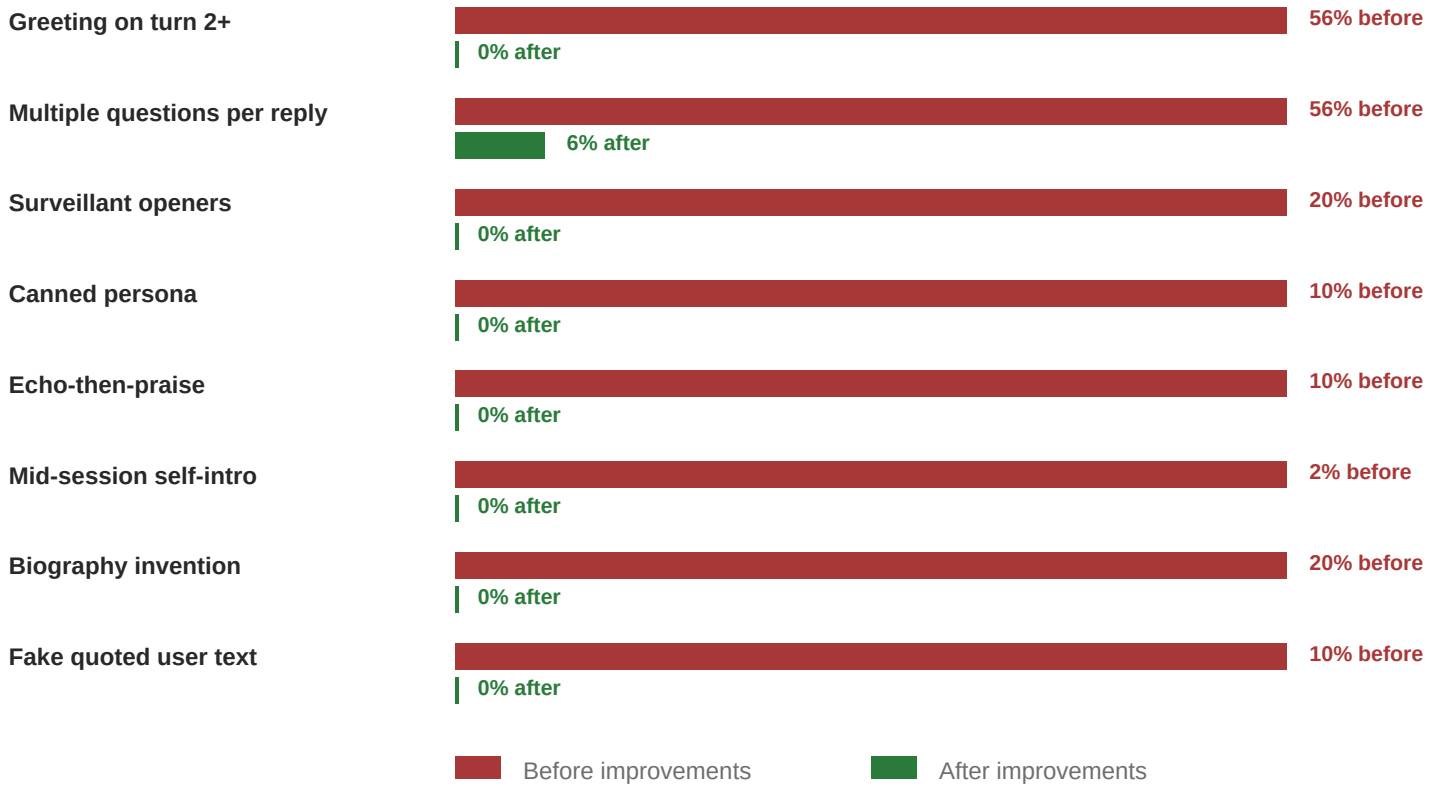
We added 12 deterministic post-processing checks that catch the specific failure patterns the prompt couldn't fully suppress - things like greeting on every turn, stacking too many questions, inventing personal anecdotes.

Outcome.

Across the 8 quality issues we tracked, 7 went to 0% violation rate and the eighth (too many questions per reply) dropped from 56% of sessions affected to just 6%. Total violation count fell from 229 to 5. Average reply length tightened to 25.7 words (down from 31.5), making conversations crisper without losing warmth.

Quality issues - before vs after

Each row shows the percentage of the 50 sessions that contained at least one instance of that quality issue. Bars are scaled to the baseline value.



By the numbers

Metric	Before	After	Change
Total Maya replies analysed	427	432	+5
Average reply length (words)	31.5	25.7	-5.8
Average turns per session	8.5	8.6	+0.1
Total quality violations	229	5	-224
Violations per reply (rate)	53.6%	1.2%	-52.5 pp
Sessions with at least one violation (worst category)	28 / 50	3 / 50	-25 sessions
Categories at 0% violation rate	0 / 8	7 / 8	+7 categories

Test coverage

Pattern	Sessions	Profile	What it stresses
Same-day back-to-back sessions (Tier A)	10	Priyansh, all turns same day	Tests intra-day repetition + opener variation
Consecutive-day sessions (Tier B)	15	Aarti, days 0-14	Tests memory accumulation + mood arc + c
Sporadic-gap sessions (Tier C)	10	Rohan, gaps of 5-12 days	Tests returning-after-pause behaviour
Cold-start single sessions (Tier D)	10	10 different fresh users	Tests first-session onboarding behaviour
Weekly deep-run sessions (Tier E)	5	Neha, 5 weeks	Tests long-term consistency + skill curriculum

Diversity in the test pool

- 10 distinct synthetic user profiles
- 8 Indian regional mother tongues represented (Hindi, Marathi, Bengali, Tamil, Telugu, Punjabi, Gujarati, Kannada)
- 5 English fluency levels covered (A1 to C1, beginner to fluent)
- 3 engagement styles tested (terse, medium, chatty)
- 12 emotional states sampled per session (tired, anxious, excited, lonely, hopeful, etc.)
- 32 days of simulated calendar time across the 50 sessions

Issue-by-issue detail

Per quality issue: total violations, sessions affected, and the reduction achieved. Sessions = 50 in each run.

Issue	Before #	After #	Before %sess	After %sess	Before rate	Reduction
Greeting on turn 2+	131	0	56%	0%	30.7%	-100%
Multiple questions per reply	53	5	56%	6%	12.4%	-89%
Surveillant openers	12	0	20%	0%	2.8%	-100%
Canned persona	6	0	10%	0%	1.4%	-100%
Echo-then-praise	5	0	10%	0%	1.2%	-100%
Mid-session self-intro	1	0	2%	0%	0.2%	-100%
Biography invention	16	0	20%	0%	3.7%	-100%
Fake quoted user text	5	0	10%	0%	1.2%	-100%
Total violations	229	5	-	-	53.6%	-98%

Read this table left-to-right per row: # = total violation count across all 50 sessions. % sess = the share of the 50 sessions that contained at least one violation of that type. Rate = violations as a percentage of all Maya replies (427 in before, 432 in after). Reduction = relative drop in % sessions affected.

Methodology

How we tested

We built 10 synthetic test users covering diverse profiles (Hindi, Tamil, Bengali, Marathi, Punjabi, Gujarati, Telugu, Kannada speakers; A1 through C1 English fluency; various professions and emotional states). Each user was simulated by a separate language model playing the user side, so we could run repeatable conversations.

The 50-session pattern

Across the 50 sessions we covered: 10 same-day back-to-back sessions (testing intra-day repetition), 15 consecutive-day sessions (testing memory accumulation across two weeks), 10 sporadic-gap sessions (testing returns after long pauses), 10 cold-start single sessions (testing first-time onboarding), and 5 weekly deep-runs over a month (testing long-term consistency). The same schedule was used for both the before and after runs.

What we measured

Eight specific quality issues that show up when AI chat assistants feel scripted or robotic: re-greeting on every turn, asking too many questions at once, surveillant openers, recycling the same character details, quiz-style English grading, mid-session self-introductions, inventing personal anecdotes, and quoting the user with words they never actually said.

The two improvements applied

First, the prompts were re-written specifically for the Qwen 32B model behind Maya - tighter constraints, explicit forbidden phrasings, and a structured reply template. Second, we added 12 deterministic post-processing checks that scan Maya's reply before the user sees it and strip specific violation patterns. The post-processing layer adds less than 5 milliseconds of latency.

What this report does NOT measure

This study covers structural quality issues that can be reliably detected. It does not measure subjective tone, the appropriateness of memory references, personality consistency over many sessions, or topic-transition smoothness. Those continue to improve through prompt iteration.